

9.4.2.2.1 Mahalanobis distance-based separability measure

$$\begin{array}{lcl} Q: & C_1 & (1,2) \quad (2,3) \quad (2,1) \\ & C_2 & (4,2) \quad (3,4) \quad (4,3) \end{array}$$

$$\text{Calculate } J_{1,2} = (m_1 - m_2)^T C^{-1} (m_1 - m_2)$$

$$\begin{aligned} \text{Solution } \textcircled{1} m_1 &= \left(\frac{1+2+2}{3}, \frac{2+3+1}{3} \right) \\ &= \left(\frac{5}{3}, 2 \right) \end{aligned}$$

$$\begin{aligned} m_2 &= \left(\frac{4+3+4}{3}, \frac{2+4+3}{3} \right) \\ &= \left(\frac{11}{3}, 3 \right) \end{aligned}$$

$$m_1 - m_2 = \left(\frac{5}{3}, 2 \right) - \left(\frac{11}{3}, 3 \right) = (-2, -1)$$

$$\textcircled{2} C = \begin{bmatrix} \text{COV}(C_1, C_1) & \text{COV}(C_1, C_2) \\ \text{COV}(C_2, C_1) & \text{COV}(C_2, C_2) \end{bmatrix}$$

$$\text{COV}(C_1, C_2) = \frac{\sum_{i=1}^n (C_{1i} - m_1)(C_{2i} - m_2)}{n}$$

$$C_1 \quad (1, 2) \quad (2, 3) \quad (2, 1)$$

$$C_2 \quad (4, 2) \quad (3, 4) \quad (4, 3)$$

$$m_1 = \left(\frac{5}{3}, 2\right) \quad m_2 = \left(\frac{11}{3}, 3\right)$$

$$C_1' \quad \left(-\frac{2}{3}, 0\right) \quad \left(\frac{1}{3}, 1\right) \quad \left(\frac{1}{3}, -1\right)$$

$$C_2' \quad \left(\frac{1}{3}, -1\right) \quad \left(-\frac{2}{3}, 1\right) \quad \left(\frac{1}{3}, 0\right)$$

$$\text{COV}(C_1, C_2) = \frac{1}{3} \times \left[-\frac{2}{3} \times \frac{1}{3} + 0 \times (-1) + \frac{1}{3} \times \left(-\frac{2}{3}\right) + 1 + \frac{1}{9} \right]$$

$$= \frac{1}{3} \times \left[-\frac{2}{9} - \frac{2}{9} + 1 + \frac{1}{9} \right]$$

$$= \frac{1}{3} \times \left(-\frac{4}{9} + \frac{10}{9} \right)$$

$$= \frac{2}{9}$$

$$\begin{aligned}
 \text{COV}(C_1, C_1) &= \frac{1}{3} \times \left[\frac{4}{9} + \frac{1}{9} + 1 + \frac{1}{9} + 1 \right] \\
 &= \frac{1}{3} \times \left[\frac{6 + 2 \times 9}{9} \right] \quad 24 \\
 &= \frac{8}{9}
 \end{aligned}$$

$$\begin{aligned}
 \text{COV}(C_2, C_2) &= \frac{1}{3} \times \left[\frac{1}{9} + 1 + \frac{4}{9} + 1 + \frac{1}{9} \right] \\
 &= \frac{1}{3} \times \left[\frac{6 + 2 \times 9}{9} \right] \\
 &= \frac{8}{9}
 \end{aligned}$$

$$C = \begin{bmatrix} \frac{8}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{8}{9} \end{bmatrix}$$

$$J_{12} = (m_1 - m_2)^T C^{-1} (m_1 - m_2)$$

$$= (-2, -1) \begin{bmatrix} \frac{8}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{8}{9} \end{bmatrix}^{-1} \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$$= \frac{24}{5}$$

$$= 4.8$$